place on the near side, very close to the equatorial region. This narrow range of sample areas doesn't provide scientists with the necessary variety of soil and rock samples to form a sound scientific theory. The small samples obtained from probes and spectral imaging shows that the Moon is very geologically diverse and in studying the diversity can the origins of the Moon be truly discovered.

As of now we can only guess at the origins from general observations.

Giant Impact or Isotopic Twin or The Big Splat

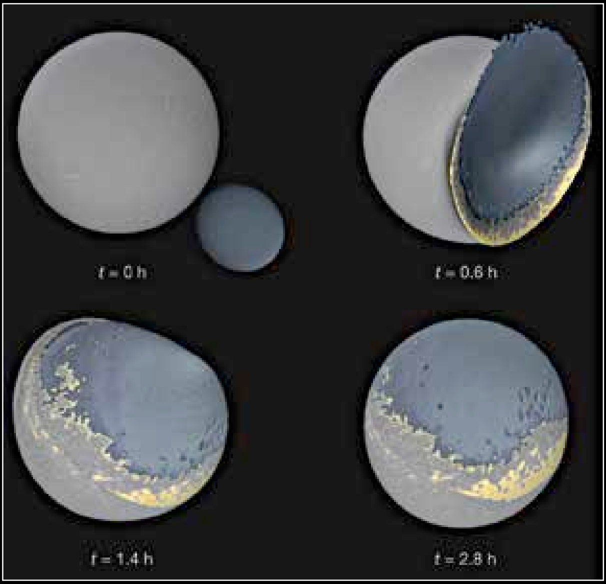
The most popular theory amongst lunar scientist is that about 4.5 billion years ago a wandering Mars-sized planet (called Theia) collided with the young Earth and resulted in the formation of massive debris, that eventually coalesced into the Moon. And although this theory has had a significant following it is still ill equipped to answer certain inconsistencies. A major one being the fact that the Earth and the Moon have an odd variety of isotopic (sharing common element isotopes) similarities in their compositions while the Moon doesn't conform to most other objects in the Solar System.

To answer these inconsistencies the theory of Isotopic Twins was floated, which sought to fill in the blanks. If the Moon was a result of an impact between the Earth and Theia, then the Moon would display isotopic features of both Earth and a cosmic body, but it doesn't, so its more likely that the Moon is a dwarf Earth twin. However, even that theory isn't totally verifiable since the only samples that show isotopic similarities were taken from the surface of the Moon. In order to truly prove a cosmic connection samples from the mantle of the Moon would be necessary.

A third theory explains more than just these inconsistencies and also addresses the strange differences between the surface of the near and far side of the Moon. The Moon's near side, as we know, always faces us and appears relatively smooth - which is a result of hardened ancient lava not found on the far side. The far side on the other hand is highly asymmetrical, filled with highlands and craters across its surface.

Scientists theorise that although the Moon was created due to a large impact from a planet sized body with the Earth, another smaller Moon was also created. A subsequent low velocity collision between our Moon and the smaller Moon resulted in what is known as - The Big Splat.

Using computer modelling and simulations, scientists at the University of California have proposed that the smaller moon's crash with our Moon is what accounts for the distinctive nature of the dark side. The low velocity collision didn't result in a large crater, instead the collision wen "splat" causing the creation of the lunar highlands.



Computer simulations show how a small second Moon from the Theia-Earth impact may be responsible for scarring the Moon's far side.

In order to verify these theories, NASA's Gravity Recovery and Interior Laboratory (GRAIL) missions are probing the Moon's gravity field to model the differences between its near and far side. The goal is to measure the gravitational effects of the materials under the crust of the Moon and test the possibility of the two Moon theory. Only time and commitment will tell which one of these theories is correct.

Where We're Going, We Don't Need Roads

Soon after the Soviet's had launched Sputnik and gained an advantage in the space race America was brimming with competition. To address the tension, on a hot summer's evening in May 1961, the President of the United States - John F. Kennedy - stood before the American public and congress, and made one of the boldest and most unsubstantiated claims ever. He promised that the United States would put a man on the Moon within ten years. They did it in 8.

The strangest and perhaps greatest aspect of the humans spirit can be analogised in that instance. The sheer human audacity to attempt the seemingly impossible given the relative ignorance of the challenge. At the time of his announcement, President Kennedy and the United States had neither the technology nor the research to accomplish the goal of landing a man on the Moon and bringing him back to Earth safely. And yet, the impossible was accomplished.

It almost feels as if the times when humanity did large, great things is diminishing. No longer do we strive for the impossible or yearn for dreams that have no practical value. The space age was actually an arms race in disguise but it brought forth an era of innovation that led to the prosperity of the whole human race; from satellite technology, telecommunications, cell phones, agriculture research, weather system analysis and so much more. All because one man made one stupid promise to follow a dream and a nation followed him there.

India is well on its way to Mars with the Mangalyaan probe but that journey too has been difficult, if not in its science then in its politics. There is a time and a place for innovation and history tells us there is nothing more ambitious than to understand the nature of the universe. So, perhaps it's time for India to get involved and send a manned mission to the Moon with the agenda to finally unveil the mystery of this distant lover once and for all. d